



RESEARCH ARTICLE

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Key Points:

- Water retention curve becomes steeper with increasing hydrate saturation
- Relative water and gas permeability is affected by hydrate saturation and morphology
- Brooks-Corey model predicts permeability very well for heterogeneously distributed hydrate system

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The water retention curve and relative permeability for gas production from hydrate-bearing sediments: pore-network model simulation

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Abstract The water retention curve and relative permeability are critical to predict gas and water production from hydrate-bearing sediments. However, values for key parameters that characterize gas and water flows during hydrate dissociation have not been identified due to experimental challenges. This study utilizes the combined techniques of micro-focus X-ray computed tomography (CT) and pore-network model simulation to identify proper values for those key parameters, such as gas entry pressure, residual water saturation, and curve fitting values. Hydrates with various saturation and morphology are realized in the pore-network that was extracted from micron-resolution CT images of sediments recovered from the hydrate deposit at the Mallik site, and then the processes of gas invasion, hydrate dissociation, gas expansion, and gas and water permeability are simulated. Results show that greater hydrate saturation in sediments lead to higher gas entry pressure, higher residual water saturation, and steeper water retention curve. An increase in hydrate saturation decreases gas permeability but has marginal effects on water permeability in sediments with uniformly distributed hydrate. Hydrate morphology has more significant impacts than hydrate saturation on relative permeability. Sediments with heterogeneously distributed hydrate tend to result in lower residual water saturation and higher gas and water permeability. In this sense, the Brooks-Corey model that uses two fitting parameters individually for gas and water permeability properly capture the effect of hydrate saturation and morphology on gas and water flows in hydrate-bearing sediments.

1. Introduction

The fundamental expressions for unsaturated soil behavior are anchored in capillarity and water saturation relation P_c-S_w , typically known as a water retention curve (WRC). The water retention curve essentially captures pore-scale characteristics of the porous media and the gas-fluid-mineral interactions. This curve can be used as an indicator to hydraulic conductivity [Assouline, 2001], soil-water storage [Brady and Weil, 2007], and soil stiffness, strength, and volume changes [Fredlund et al., 1996; Gens and Alonso, 1992; Öberg and Sällfors, 1997; Pedarla et al., 2012]. The relative fluxes of liquid and gaseous phases through porous media are usually captured by relative water and gas permeability k_{rw} and k_{rg} . Both the water retention curve and relative permeability are the most critical characteristics to characterize multiphase flow in porous media.

Widely used models of water retention curve and relative permeability [Bear, 1979; Bear and Cheng, 2010; Brooks and Corey, 1964; Corey, 1954; Parker et al., 1987; Stone, 1970; van Genuchten, 1980] contain multiple fitting parameters. The appropriate selection of these parameter values is critical to enhance the prediction accuracy of water and gas production rates for the development of methane gas hydrate deposits [Anderson et al., 2011; Jang and Santamarina, 2011, 2014; Johnson et al., 2011; Kleinberg et al., 2003; Li et al., 2013; Mahabadi and Jang, 2014]. However, this selection has long been a challenge due to the lack of experimental studies that can precisely control and quantify hydrate saturation throughout the tests. Reported direct measurement of relative water permeability in natural sediments is available for Nankai Trough pressure cores after hydrate dissociation [Santamarina et al., 2015]. Recently, percolation theory has been deployed to characterize relative gas and water permeability in hydrate-bearing sediments [Daigle and Rice, 2015;

Ghanbarian *et al.*, 2015]. Yet, none of these studies has been able to capture all the equations and fitting parameters for hydro-thermo-mechanically coupled process analyses in numerical simulators for gas hydrate research [Anderson *et al.*, 2011; Moridis *et al.*, 2008].

This study aims to identify proper parameter values for characterizing the water retention curve and relative permeability in hydrate-bearing sediments using the combined techniques of micro-focus X-ray computed tomography (CT) and pore-network model simulation. The sediment recovered from a hydrate deposit at the Mallik site in Canada is remolded and scanned at in situ stress conditions. A three-dimensional pore network is then extracted from the micron-resolution CT images for pore-network simulation. Hydrate is artificially generated in the pores with various saturations and morphology. The processes of gas invasion, hydrate dissociation, and gas expansion during dynamic gas production are simulated to obtain the proper fitting parameter values for water retention curve and relative permeability of water and gas in hydrate-bearing sediments.

2. Fundamentals: Water Retention Curve and Relative Permeability Models

The water retention curve is affected by pore throat size distribution, connectivity and spatial correlation, soil fabric, contact angle, and interfacial tension [Dai and Santamarina, 2013]. Obtaining the water retention curve for sediments with the presence of hydrate faces many experimental challenges, such as long induction time for hydrate formation and poor control of hydrate saturation due to transient hydrate dissolution/dissociation and formation during water and gas flow.

Analytical expressions usually capture the water retention curve using P_c , S_{rw} , and residual water saturation S_{rw} [Brooks and Corey, 1964; Corey, 1954; Fredlund and Xing, 1994; van Genuchten, 1980]. Most numerical studies on gas hydrates [Anderson *et al.*, 2011; Gamwo and Liu, 2010; Hong and Pooladi-Darvish, 2003, 2005; Moridis and Reagan, 2007; Moridis and Sloan, 2007; Reagan and Moridis, 2008; Reagan *et al.*, 2010; Rutqvist and Moridis, 2007] use the van Genuchten [1980] model:

$$P_c = P_0 \left[\left(\frac{S_w - S_{rw}}{1 - S_{rw}} \right)^{-\frac{1}{m}} - 1 \right]^{1-m} \quad (1)$$

where P_0 is the gas entry pressure, S_{rw} is the residual water saturation, and m is a fitting parameter. Lower m -value shows steeper P_c - S_w curve (Figure 1a), typically in sediments with wider pore size distribution.

Additionally, the van Genuchten [1980] model is used to predict the relative water k_{rw} and gas k_{rg} permeability [Parker *et al.*, 1987]:

$$k_{rw} = \bar{S}^{0.5} \left[1 - \left(1 - \bar{S}^{1/m} \right)^m \right]^2, \quad \bar{S} = \left(\frac{S_w - S_{rw}}{1 - S_{rw}} \right) \quad (2)$$

$$k_{rg} = (1 - \bar{S})^{0.5} \left[1 - \bar{S}^{1/m} \right]^{2m}, \quad \bar{S} = \left(\frac{S_w - S_{rw}}{S_{wmax} - S_{rw}} \right) \quad (3)$$

where S_{wmax} is the water saturation at which gas permeability starts to occur. A lower m -value depicts lower water permeability but higher gas permeability at a given saturation (Figure 1b).

An earlier relative permeability model [Brooks and Corey, 1964] also frequently used in hydrate simulation studies is:

$$k_{rw} = \left(\frac{S_w - S_{rw}}{1 - S_{rw}} \right)^{n_w} \quad (4)$$

$$k_{rg} = \left(\frac{S_g - S_{rg}}{1 - S_{rw}} \right)^{n_g} \quad (5)$$

where S_{rg} is the residual gas saturation, n_w and n_g are fitting parameters for water and gas permeability. Lower n_w and n_g values reflect higher relative water and gas permeability (Figure 1c). Note that two individual fitting parameters, n_w and n_g , are used to predict relative water and gas permeability separately in

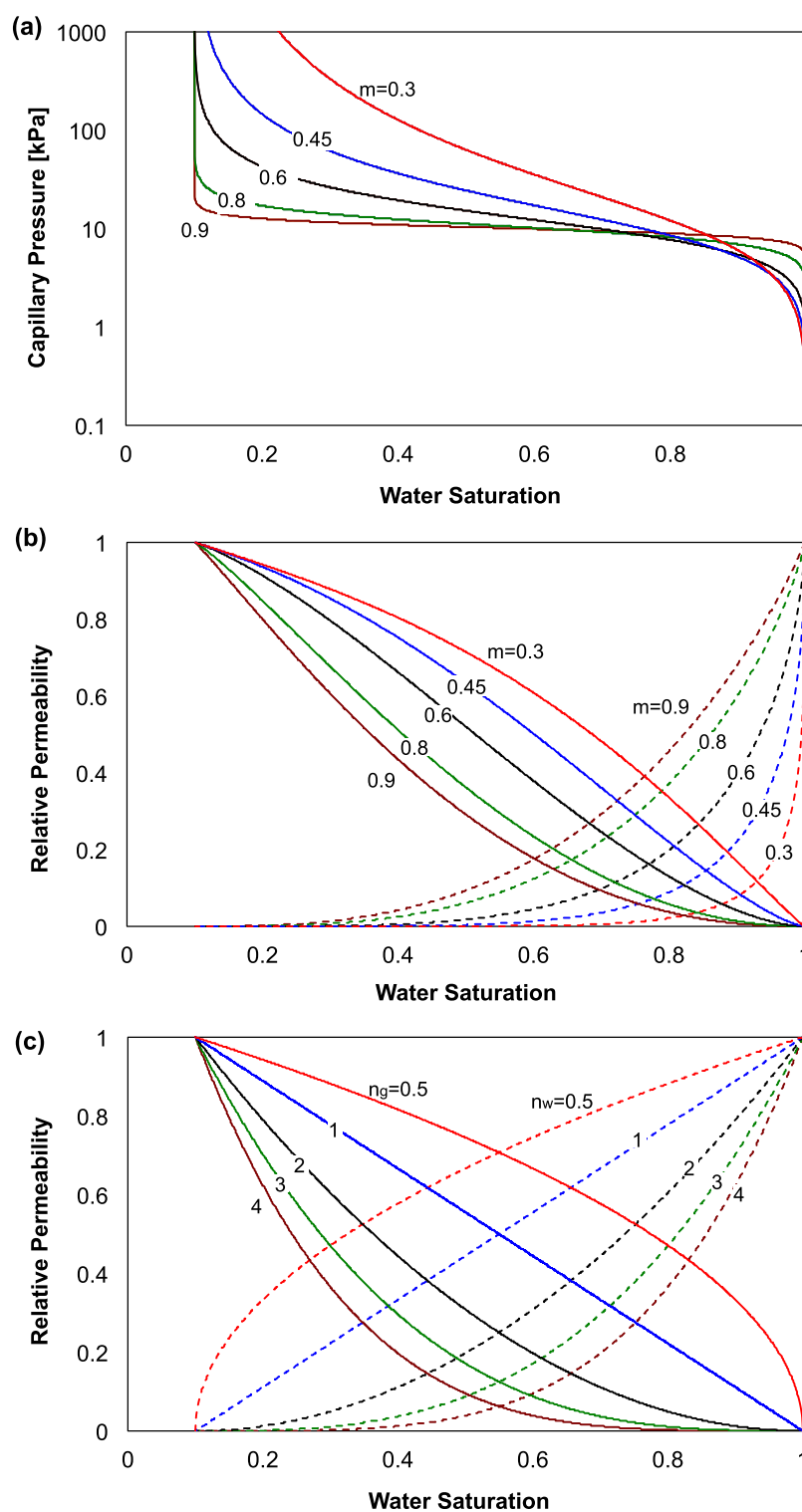


Figure 1. Effects of fitting parameters on the water retention curve (P_c -vs- S_w) and relative permeability. (a) Water retention curve using the *van Genuchten* model (equation (1)). (b) Relative permeability using the *van Genuchten* model (equations (2) and (3)). As an m -value increases, relative water permeability increases while relative gas permeability decreases. (c) Relative permeability using the Brooks-Corey model (equations (4) and (5)). As fitting parameters (n_g and n_w) increase, relative water and gas permeability decrease.

Brooks-Corey model, while one common m -value is used for both water and gas permeability equations in van Genuchten model.

3. Methodology

3.1. X-Ray CT Scanning and Pore-Network Extraction

The sediment used for X-ray scanning was recovered from Mallik 5L-38 site at 1091 m depth. A sieve analysis was conducted to obtain the grain size distribution (Figure 2a, refer also to Jenner *et al.* [1999] for the grain size distribution of other specimens at this site). This sediment was oven-dried for 48 hours, and then packed in a rubber sleeve (with an inner diameter of 6.35 mm) housed within a high-pressure vessel made out of beryllium for X-ray scanning. The in-situ effective stress (~ 10 MPa) was restored during scanning (Note that this is a remolded sample that does not preserve the original pore structure and connectivity of in-situ sediment, but may have a similar porosity due to restored confining stress). The micro-focus scanner has ~ 0.68 μm per pixel resolution with 0.7 mm field of view in all three dimensions. However, we intentionally chose a larger pixel size so the overall scanned volume is large enough to be representative. In this study, the volume of the scanned image is 27 mm^3 (3 mm $\times$ 3 mm $\times$ 3 mm) with 12.5 $\mu\text{m}/\text{pixel}$ resolution. The obtained CT images provide the three-dimensional structure of the scanned sediment, including both the grains and the pore spaces (Figure 2b). Then, a three-dimensional pore network consisting of spherical pores and cylindrical tubes (Figure 2c) is extracted from the CT images using the maximal ball algorithm developed by Silin and Patzek [2006]. The maximal ball algorithm finds spheres inscribed in the pore wall. As a general rule, big spheres located in the center of pore spaces of the sediment image become pores of

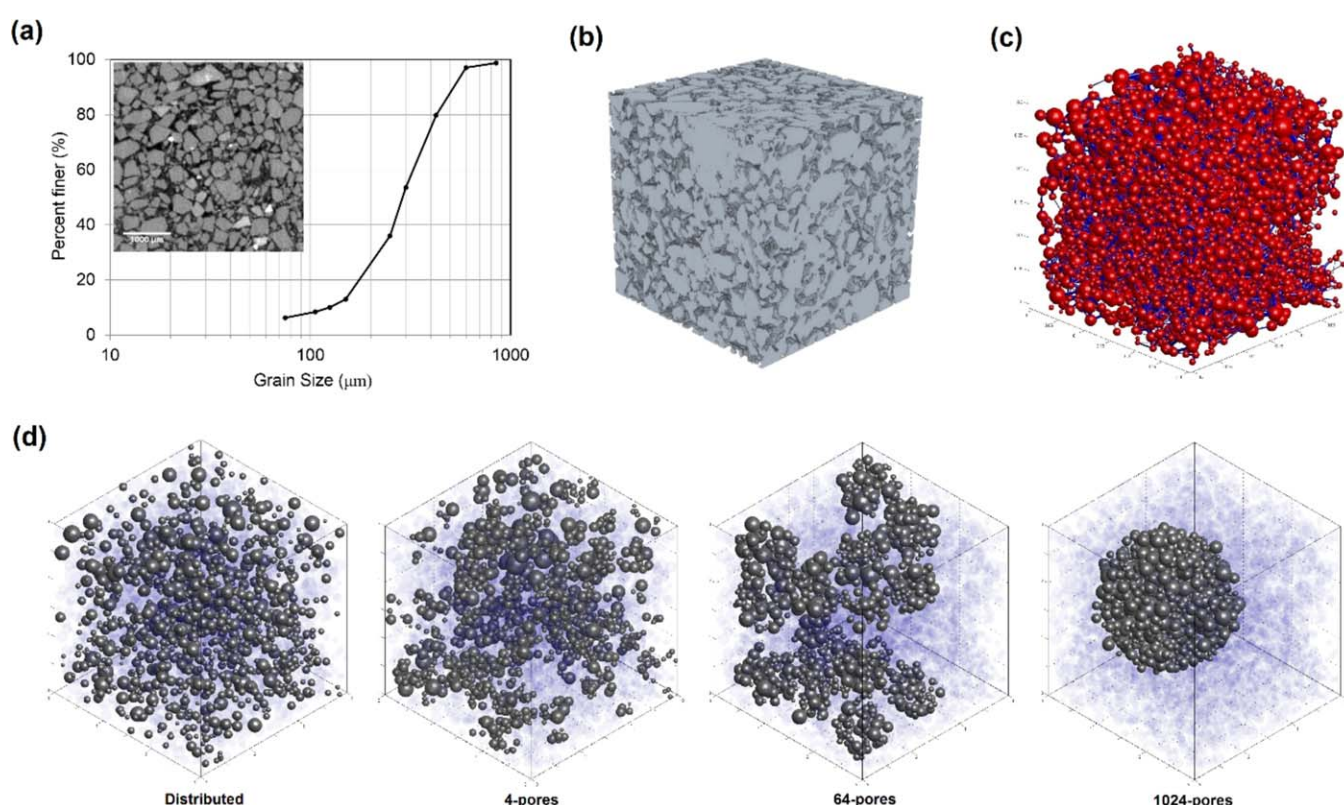


Figure 2. Pore-network model generation and hydrate distribution in the pore-network model. (a) Grain size distribution of the specimen recovered from the Mallik 5L-38 site. The soil is classified as poorly graded sand with fines based on the Unified Soil Classification System (ASTM D2487). The coefficient of uniformity is $C_u = D_{60}/D_{10} = 0.33 \text{ mm}/0.13 \text{ mm} = 2.6$ and the coefficient of curvature is $C_c = D_{30}^2/(D_{10}D_{60}) = (0.23 \text{ mm})^2/(0.13 \text{ mm} \times 0.33 \text{ mm}) = 1.2$ (D_x is the particle diameter representing that X percent of the total particles is smaller than D). The inset shows a cross section of the CT image of the Mallik sand specimen. (b) Three-dimensional representative pore space (3 mm $\times$ 3 mm $\times$ 3 mm) obtained by μCT scan. (c) Pore network model extracted from the pore space CT images. The pore network model consists of 5376 pores and 18770 tubes with a mean tube connectivity per pore (coordination number) of $cn = 6.8$. Mean pore radius is $\mu[R_{\text{pore}}] = 60 \text{ } \mu\text{m}$, mean tube size is $\mu[R_{\text{tube}}] = 11 \text{ } \mu\text{m}$, and mean tube length is $\mu[L_{\text{tube}}] = 41 \text{ } \mu\text{m}$ ($\max[R_{\text{pore}}] = 185 \text{ } \mu\text{m}$, $\min[R_{\text{pore}}] = 21 \text{ } \mu\text{m}$, $\max[R_{\text{tube}}] = 50 \text{ } \mu\text{m}$, $\min[R_{\text{tube}}] = 1 \text{ } \mu\text{m}$, $\max[L_{\text{tube}}] = 190 \text{ } \mu\text{m}$, $\min[L_{\text{tube}}] = 0.04 \text{ } \mu\text{m}$). (d) Various types of hydrate morphology (distributed versus patchy) for a given hydrate saturation $S_h = 0.2$. Hydrate forms in randomly distributed individual pores, or in patches of different size consisting of 4 pores, 64 pores, and 1024 pores. Note that hydrate pores are shown in gray and water pores in transparent blue.

the pore-network model and the small spheres inscribed in pore throats are turned into tubes. Further information on inscribing spheres and assigning spheres for pores or tubes is found in *Dong and Blunt* [2009]. The generated pore-network model can be modified to accurately simulate multiphase fluid flows by changing pore and tube size distribution or considering irregular cross-sectional shape of tubes [*Valvatne and Blunt*, 2004], but in this study, the extracted pore-network model was used without further modification. Detailed information on the extracted pore-network model is provided in the caption of Figure 2.

3.2. Hydrate Realization: Saturation and Morphology

Three types of hydrate morphology are found in natural sediments: (1) Pore-filling, (2) lenses/veins, and (3) nodules/chunks [*Waite et al.*, 2009]. The hydrate morphology is determined by the effective stress level of the natural sediments, pore and throat size, and hydrate-water interfacial tension [*Clennell et al.*, 1999; *Dai et al.*, 2012]. In our study, we consider only pore-filling model usually found in coarse-grained sediments.

To achieve target initial hydrate saturation (i.e., $S_h=0, 0.2, 0.4$, and 0.6 in this study), hydrates are assigned in randomly chosen pores, not in tubes. Hydrate pore selection disregards their size. And it is assumed that the selected pores are entirely filled with hydrates. The assumption of either 100% or 0% pore occupancy by hydrates reflects Ostwald ripening. In a long-term, Ostwald ripening in coarse-grained sediments promotes patchy hydrate formation, whereby hydrate-filled sediment patches ($S_h=100\%$) are embedded in hydrate-free sediments ($S_h=0\%$). Reported data (e.g., P-wave velocity and hydraulic conductivity at different hydrate saturation) and physical analyses suggest hydrate-bearing sands contain a heterogeneous and patchy hydrate distribution [*Dai et al.*, 2012]. And the laboratory observation performed over two weeks for THF hydrate confirms the formation of hydrate patch [*Mahabadi et al.*, 2016].

Based on the fact that hydrate patch size and morphology affects the physical properties of hydrate-bearing sediments in orders of magnitude [*Dai and Santamarina*, 2013; *Dai et al.*, 2012; *Yun et al.*, 2005], various hydrate morphology are considered in this study: (1) hydrates in individually distributed pores (Figure 2d) and (2) hydrates in patchy formations of different sizes, occupying 4, 64, or 1024 pores for a given hydrate saturation (Figure 2d). The size of hydrate patch is arbitrarily chosen to explore the effect of hydrate patch size on water retention curve and relative permeability.

Due to hydrate occupancy in pores, the pore connectivity of pore-network model is reduced, which affects residual water saturation as well.

3.3. Water Retention Curves in Hydrate-Bearing Sediments

The water retention curve is obtained by simulating the gas invasion process into water-saturated, hydrate-bearing sediments. Gas invasion is enforced at tubes located on the inlet boundary. Quasi-static gas invasion controlled by capillarity is assumed so the viscosity effect can be disregarded. If the gas pressure is higher than the summation of the capillary and water pressure of the water-filled tube ($P_g > P_w + P_c$), the gas can invade the tube. The capillary pressure at gas-water interface in a tube with radius R_{tube} is captured by the Laplace equation $P_c = 2T_s \cos \theta / R_{tube}$, where in this study the water-gas interfacial tension is $T_s = 0.072$ N/m; the contact angle is assumed $\theta = 0^\circ$ to represent a completely water-wet surface (note: refer to *Al-Futaisi and Patzek* [2003] for the effects of contact angle on computed relative permeability). With pressure increases, gas starts invading through the largest tube on the inlet boundary. And as capillary pressure increases, gas invades more water-filled tubes, leading to a decrease in water saturation; thus, the water retention curve is computed until water drainage stops.

3.4. Relative Permeability After Hydrate Dissociation

Solid hydrate turns into gas and water during hydrate dissociation. Hydrate dissociation and the volume expansion of the dissociated gas is controlled by gradually lowering the pressure at pore-network inlet and outlet boundaries from $P=15$ MPa to 0.1 MPa at constant $T=287$ K. The gas volume expansion upon dissociation and during depressurization is computed by the modified Peng-Robinson equation of state [*Jang and Santamarina*, 2011; *Stryjek and Vera*, 1986]. The depressurization is kept slow enough so that the endothermic effect during hydrate dissociation is negligible. As soon as the P-T condition comes out of the hydrate phase boundary ($P=13.9$ MPa and $T=287$ K), hydrates start to dissociate. The water from dissociated hydrate is drained out of the pore-network (for all cases, water pores are forming percolating path), but the gas from dissociated hydrates remains in the pore space that was initially occupied by hydrates. At the P-T condition of $P=13.9$ MPa and $T=287$ K, the volume of the gas dissociated from hydrates is the same as

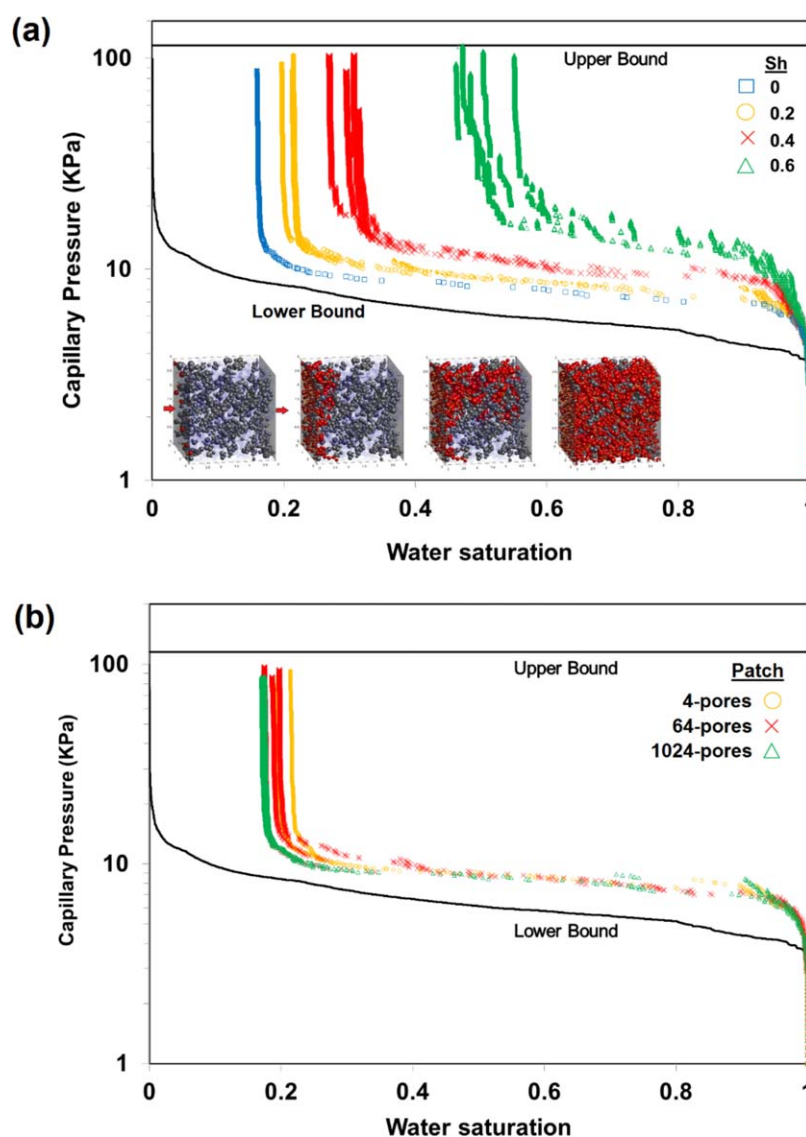


Figure 3. Water retention curves in hydrate-bearing sediments. (a) Effects of hydrate saturation ($S_h=0, 0.2, 0.4$, and 0.6) on computed water retention curves. Upper and lower bounds are obtained by injecting gas into tubes arranged in a series forming a single line from smallest to largest (upper bound) or in parallel (lower bound). Inset figures show gas invasion into the hydrate-bearing pore-network from left to right boundary. Note that gas pores are colored red, hydrate pores gray, and water pores transparent blue. (b) Effects of hydrate morphology (all at constant hydrate saturation $S_h=0.2$) on water retention curves.

the volume of initial hydrates. Therefore, pores occupied by initially hydrates turn into gas pores (for $S_h=0.2$ case, the saturations for hydrate, gas, and water prior to dissociation are $S_h=0.2$, $S_g=0$, $S_w=0.8$, respectively, and become $S_h=0$, $S_g=0.2$, $S_w=0.8$, right after dissociation at $P=13.9$ MPa and $T=287$ K). Then, further depressurization allows gas to expand into neighboring water pores. Gas expansion continues as long as gas pressure is higher than the summation of capillary pressure and water pressure, and as long as water drainage path is available. At around water saturation $S_w=0.15\sim 0.2$, gas expansion stops mainly due to the loss of water drainage path.

The methane solubility in water without hydrate is assumed zero due to relatively small methane solubility in water compared to large methane amount contained in methane hydrates. The gas expansion algorithm and water and gas permeability calculation during each gas expansion step are explained in detail in the authors' previous publication [Jang and Santamarina, 2011; Jang et al., 2011; Mahabadi and Jang, 2014].

4. Simulation Results

4.1. Computed Water Retention Curves

Figure 3a shows the computed retention curves in sediments with various initial hydrate saturations, i.e., $S_h=0, 0.2, 0.4$ and 0.6 . Hydrates (black spheres Figure 3a, inset) are randomly distributed in individual pores. One simulation run is performed for $S_h=0$ case and five simulation runs for each of other $S_h>0$ cases. All cases use the identical pore-network. The process of gas invasion into the hydrate-bearing pore network model is illustrated as red spheres in the Figure 3a inset. As the initial hydrate saturation increases, the gas-entry pressure increases, and the capillary pressure at a given water saturation also increases. The higher initial hydrate saturation condition results in the higher residual water saturation.

Figure 3b shows the computed water retention curves in sediments with the same initial hydrate saturation ($S_h=0.2$) but different hydrate patch size (i.e., the hydrate patch is composed of 4, 64, and 1024 pores). Three simulation runs are performed for each patch size and the spatial distribution of hydrate patch varied at each simulation. Sediments with larger hydrate patches tend to result in lower gas-entry pressure and also slightly lower residual water saturation (Figure 3b).

4.2. Relative Permeability During Gas Expansion

Figure 4 illustrates the gas expansion and water drainage processes during gas expansion in sediments with initial hydrate saturation $S_h=0.2$ ($S_h=0$ and $S_w=0.8$). All the hydrates dissociate into gas ($S_g=0$ and $S_w=0.8$, first column in Figure 4). And gas expands with further depressurization (second, third, and fourth columns in Figure 4). With further gas expansion, water continues draining, but gas conductivity does not occur until the gas phase forms a percolation path at gas saturation $S_g=0.3$ (second column in Figure 4). Later, the water phase gradually loses percolation paths, and the gas phase builds more gas flow paths (third column in Figure 4a). Eventually, gas keeps expanding into neighboring water-filled pores until water drainage stops at $S_g=0.86$ and $S_w=0.14=S_{wv}$ (fourth column in Figure 4a).

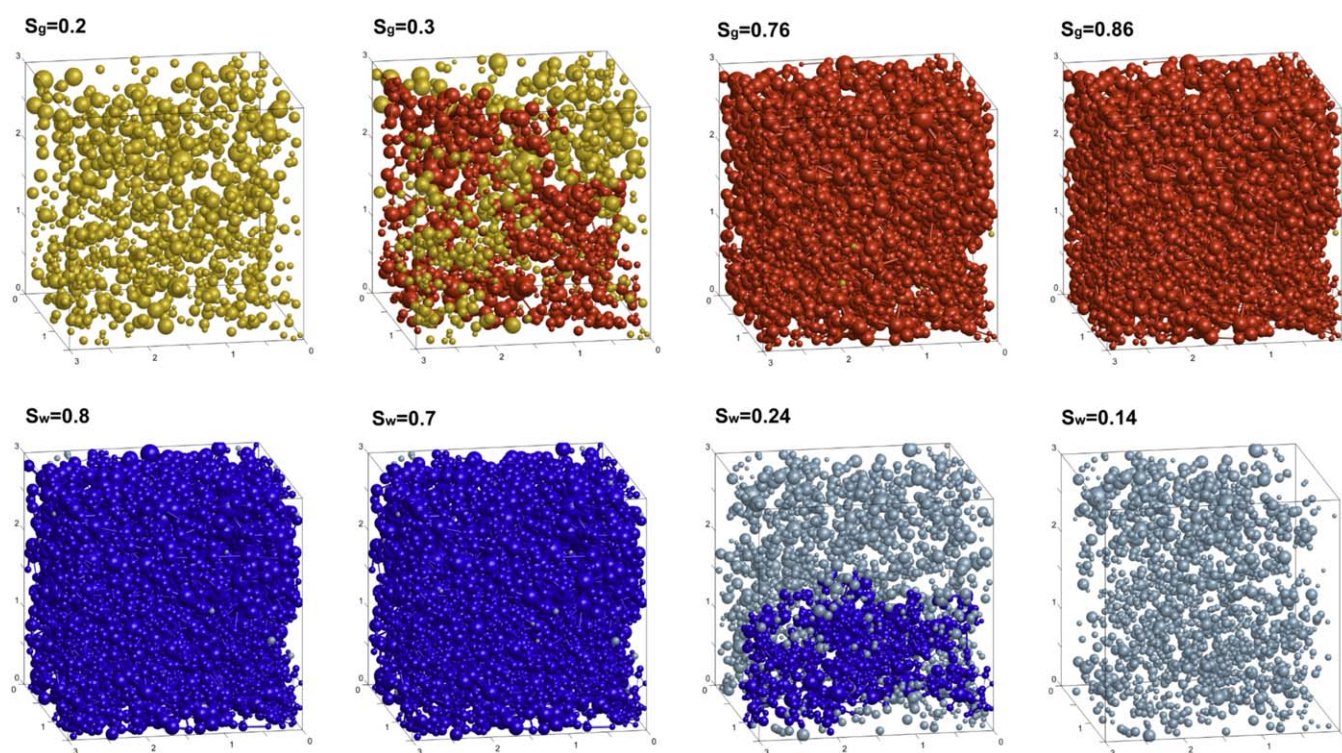


Figure 4. Illustrations of gas and water flow processes during gas expansion in sediment undergoing hydrate dissociation ($S_h=0.2$). Note that gas pores forming conductivity paths are colored red, and isolated gas pores are colored yellow in upper figures. Water pores forming conductivity paths are colored blue, and isolated water pores are colored light blue in lower figures.

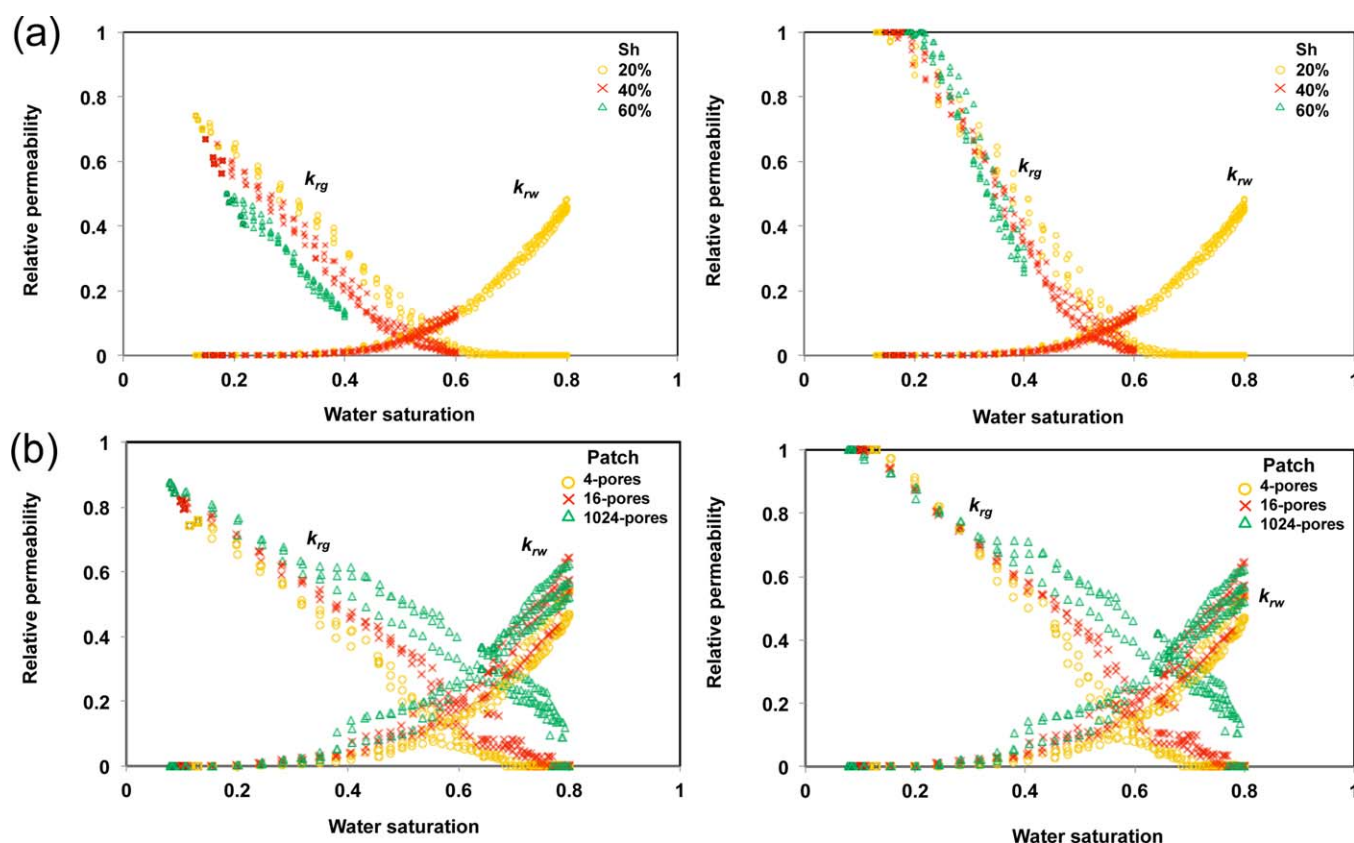


Figure 5. Relative permeability in hydrate-bearing sediments. (a) Effect of hydrate saturation on relative gas and water permeability for initial hydrate saturations $S_h = 0.2$ (yellow), 0.4 (red) and 0.6 (green). Five simulation runs are performed for each hydrate saturation condition. (b) Effect of hydrate patch size on relative gas and water permeability. Three simulation runs are performed for each size of hydrate patch: 4 pores (yellow), 64 pores (red) and 1024 pores (green). The hydrate saturation is $S_h = 0.2$. Note that in left figures, the water (or gas) conductivity is normalized by the water (or gas) conductivity at 100% water (or gas) saturation, however, the gas conductivity at the residual water saturation is used to normalize gas conductivity in right figures.

The relative permeability of water (or gas) is the conductivity at a given water saturation normalized by the water (or gas) conductivity at 100% water (gas) saturation. The gas conductivity at the residual water saturations can be also used for normalization.

In order to explore the effect of hydrate saturation and patch formation on relative water and gas permeability, the water (or gas) conductivity at a given water saturation is normalized by the water (or gas) conductivity at 100% water (or gas) saturation (Figures 5a and 5b, left). However, in order to fit van Genuchten and Brooks-Corey equations, the gas conductivity at the residual water saturation is used for normalization (Figures 5a and 5b, right).

Results show that relative gas permeability increases, and relative water permeability decreases as gas expands into water-filled pores (Figure 5). As shown in left figure in Figure 5a, sediments with lower initial hydrate saturation result in (1) higher relative gas permeability and (2) lower residual water saturation. This trend of relative gas and water permeability as a function of initial hydrate saturation is consistent with published results [Mahabadi and Jang, 2014]. Higher gas connectivity developed for lower hydrate saturation conditions can explain this trend.

Additionally, with the increase in hydrate patch's size, both water and gas permeability increases dramatically particularly for $S_w > 0.4$ (Figure 5b, left). When the water saturation is lower than $S_w = 0.4$, the effect of hydrate patch size on relative gas and water permeability becomes insignificant. Moreover, larger patch size results in lower residual water saturation.

Gas conductivity values are normalized by the gas conductivity at the residual water saturation for each simulation run in order to fit the van Genuchten equation and Brooks-Corey equation (Figures 5a and 5b, right).

Table 1. Fitting Parameters for Water Retention Curves and Relative Permeability.^a

Numerical Results	van Genuchten Model							Brooks-Corey Model		
	Water Retention Curve			Relative Permeability						
	P_0 [kPa]	S_{rw}	m	S_{rw}	S_{wmax}	m_g	m_w	S_{rw}	n_g	n_w
0	8.0	0.16	0.91							
0.2	8.7 ± 0.2	0.21 ± 0.01	0.90 ± 0.01	0.14 ± 0.02	0.80	0.87 ± 0.00	0.88 ± 0.01	0.14 ± 0.02	2.6 ± 0.1	3.1 ± 0.1
0.4	10.2 ± 0.4	0.29 ± 0.02	0.87 ± 0.01	0.17 ± 0.01	0.75	0.85 ± 0.01	0.85 ± 0.01	0.17 ± 0.01	3.1 ± 0.1	3.3 ± 0.1
0.6	13.8 ± 0.5	0.49 ± 0.04	0.81 ± 0.01	0.20 ± 0.01	0.60	0.80 ± 0.01	0.80 ± 0.02	0.20 ± 0.01	3.5 ± 0.1	3.3 ± 0.2
Patch size (at $S_h = 0.2$)										
4-pore	8.6 ± 0.2	0.20 ± 0.01	0.91 ± 0.01	0.12 ± 0.01	0.80	0.70 ± 0.01	0.89 ± 0.01	0.12 ± 0.01	2.1 ± 0.1	3.1 ± 0.2
64-pore	8.4 ± 0.2	0.19 ± 0.01	0.91 ± 0.01	0.10 ± 0.00	0.80	0.55 ± 0.02	0.97 ± 0.02	0.10 ± 0.00	1.6 ± 0.1	2.6 ± 0.2
1024-pore	8.0 ± 0.2	0.17 ± 0.00	0.92 ± 0.01	0.08 ± 0.00	0.80	0.25 ± 0.03	1.07 ± 0.02	0.08 ± 0.00	1.1 ± 0.2	2.3 ± 0.2
Recommendations				Uniformly Distributed Hydrate		Note: Impacts of Heterogeneously Distributed/Patchy Hydrate				
van Genuchten model	WRC	P_0	$P_0^{HBS} = P_0^{HFS} (1 + S_h)$			Lower P_0				
		S_{rw}	$S_{rw}^{HBS} = S_{rw}^{HFS} (1 + 5S_h)$			Lower S_{rw} with increasing patch size				
		m	$m^{HBS} = m^{HFS} (1 - 0.15S_h)$			Negligible impact when $S_h = 0.2$				
	Relative Permeability	S_{rw}	$S_{rw}^{HBS} = 0.1 + 0.17S_h$			Lower S_{rw}				
		S_{wmax}	$S_{wmax}^{HBS} = 1 - 0.7S_h$			No impact				
		m_g m_w	$m^{HBS} = m^{HFS} (1 - 0.19S_h)$			Significantly lower m_g , decreasing with increasing patch size Higher m_w increasing with increasing patch size				
B-C model	Relative Permeability	S_{rw}	$S_{rw}^{HBS} = 0.1 + 0.17S_h$			Lower $S_{rw} = \sim 0.1$				
		n_g	$n_g^{HBS} = 2 + 2.6S_h$			Significantly lower n_g , $n_g^{HBS} = 1 + 2.6S_h$				
		n_w	$n_w^{HBS} = \sim 3 - 3.5$			Lower n_w , $n_w^{HBS} = \sim 2 - 3$				

^aNote: HBS = hydrate-bearing sediments; HFS = hydrate-free sediments..

5. Analyses and Discussions

5.1. Pore-Network Model Simulation—Relevance to Experimental Tests

The shape of pore geometry in real sediments is irregular, while the pore-network model composed of cylindrical tubes and spherical pores simplifies the real pore geometry of sediments, which can result in differences between numerical simulation and experimental study. For example, the maximum ball algorithm used for pore-network model extraction finds spheres inscribed onto the pore surface [Dong and Blunt, 2009]. Therefore, the cross-sectional area of tubes tends to be smaller than the actual cross-sectional area of real pore throats. The simulation results obtained by using the Lattice Boltzman Method show that the capillary pressure of the cylindrical tube inscribed in other tubes with polygonal cross-sectional shape is smaller than the capillary pressure of polygonal shape tubes [Mahabadi et al., 2015]. In addition, the process of finding inscribed spheres for various pore throat shapes may result in homogenization of pore throat size distribution of real sediments. Therefore, water retention curves obtained from the pore-network model may show more uniform pore throat size distribution, which increases m-value in the P_c -vs-saturation curve, as shown in Figure 1a.

5.2. Flow in Hydrate-Bearing Sediments—Recommended Parameter Values

The method of least squares is used to fit the equations (1)–(5) to the water retention curves and relative permeability simulation results. When the least square method is applied to the unevenly distributed data points, biased fitting parameters can be obtained. Therefore, a median curve is found for distributed data. Then, evenly distributed data points on the median curve are used for the least squares method to find fitting parameters with a minimum L2-norm. Fitting parameters are calculated at each individual simulation run for each initial hydrate saturation condition and each hydrate morphology case. Then, the averaged fitting parameters of 5 or 3 simulation runs are obtained.

Table 1 summarizes the best fitting values for the van Genuchten model and the Brooks-Corey model based on the simulations. The results of water retention curve simulation suggest that the three parameters in the van Genuchten WRC model (i.e., P_0 , S_{rw} , and m) depend on hydrate saturation: increased hydrate saturation in sediments leads to (1) higher gas entry pressure P_0 , (2) higher residual water saturation S_{rw} , and (3)

steeper P_c - S_w curve (meaning lower m value). For patchy hydrate formation, as the size of hydrate patch increases for a given hydrate saturation, the m -value increases slightly (from $m=0.91$ to 0.92), and the gas-entry pressure decreases (Table 1). The gas-entry pressure for the largest hydrate patch case (a patch consisting of 1024 pores) is $P_0=8.0$ kPa which is as low as the gas-entry pressure of the hydrate-free sediment. It is obvious that sediments with heterogeneous hydrate distribution tend to have lower gas entry pressure than those with uniformly distributed hydrate.

The relative water and gas permeability during gas expansion also depend on both hydrate saturation and hydrate morphology. During hydrate dissociation, gas expands into the neighboring water-filled pores and forms interconnected clusters, which further interconnect to each other to become larger gas clusters and eventually percolate through the porous media. In sediments with distributed hydrate, the gas permeability of the sediment with higher S_h is lower than that of the sediment with lower S_h (Figure 5a), due to less interconnected gas clusters in a higher S_h case than that in a lower S_h case. Hydrate morphology has even more significant effect on the gas permeability (Figure 5b). At a given saturation, the relative gas permeability increases dramatically as the size of hydrate patch increases, e.g., $k_{rg} = \sim 0.5$ in the case of a larger hydrate patch (1024-pore size), but $k_{rg} = \sim 0.1$ in the case of a smaller hydrate patch (4-pore size) at around $S_w=0.55$. However, the variance in gas permeability due to hydrate morphology diminishes toward residual water saturation, i.e., at gas saturation $S_g > \sim 0.7$ or water saturation $S_w < \sim 0.3$.

With regard to water permeability, initial hydrate saturation in sediments with uniformly distributed hydrate has marginal effect. But in sediments with a patch hydrate, larger hydrate patch results in higher relative water permeability because gas expansion from larger hydrate patches has less effect on plugging water paths than does a smaller hydrate patch or distributed hydrates.

More importantly, the simulation results also suggest that using one common fitting parameter m -value in the van Genuchten model for both gas and water permeability is not appropriate to characterize the water/gas flows during gas production from hydrate deposits. In sediments with uniformly distributed hydrate, the m -value for both gas and water permeability decreases with increasing hydrate saturation. Under identical hydrate saturation ($S_h = 0.2$ in this case, Table 1) but various hydrate morphology, however, the m -values for gas and water permeability show evident discrepancies; and sediments with larger hydrate patches tend to result in lower residual water saturation S_{rw} and higher gas and water permeability (Table 1). In this sense, the Brooks-Corey model that has two different fitting parameters specifically for gas n_g and water n_w permeability appropriately predict the flows in sediments with patchy hydrate. The fitting parameter values for the Brooks-Corey model for sediments with various hydrate saturation and morphology are listed in Table 1.

6. Conclusions—Recommendations

Based on the results and analyses in this study, we propose the following recommendations for selecting proper parameters and their values (summarized also in Table 1) when characterizing gas and water flows in sediments during hydrate dissociation:

1. For water retention curves in hydrate-bearing sediments, the air-entry pressure P_0 , residual water saturation S_{rw} , and m -value of the van Genuchten model depend on hydrate saturation S_h . The effect of hydrate patch size on the WRC steepness m is negligible, particularly at $S_h=0.2$.
2. In relative permeability characterization using the van Genuchten model, the residual water saturation S_{rw} and maximum water saturation S_{wmax} should be corrected by considering hydrate saturation S_h . Also, the same m -value can be used for both gas and water permeability, but only in sediments with uniformly distributed hydrate; hydrate morphology greatly impacts the m -value for gas permeability.

The Brooks-Corey model characterizes the gas and water permeability using two separate parameters. Hydrate saturation slightly affects the fitting parameter for water permeability; but the gas permeability must be corrected at different hydrate saturation S_h . Again, hydrate morphology dramatically affects the gas permeability. In particular, a more heterogeneous specimen (with a larger hydrate patch) tends to result in both higher gas and water permeability.

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